Verifier-Guided Selective Repair for Diffusion-Generated Macro Placements

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Abstract

Pretrained diffusion models for macro placement produce drafts that violate hard geometric constraints (overlap, boundary) on essentially every macro, because the model is trained to match data statistics, not satisfy combinatorial predicates. We propose **Verifier-Guided Selective Repair (VSR)**: a verifier emits per-macro severity, a pairwise overlap graph, and boundary protrusions; an offender mask + a 100-step force integrator parametrized by a single wirelength/legality knob λ moves the violating macros while preserving the rest. VSR is a λ-controlled *family* of repair operators rather than a single setting; we report two reference points (λ=2, λ=8) and the full sweep. We position VSR as a repair *layer*, not a legalizer: across 6 ISPD2005 circuits and 4 seeds, no method we test (including ChipDiffusion's 5000-step legalizer and a downstream `VSR`→`cd-sched` pipeline) achieves a fully legal placement on any of the 24 trials. Within the macros-only operating point, VSR reduces residual violations by a median -55% at λ=2 and -52% at λ=8 relative to the diffusion draft; strict Pareto improvement is $12/24$ at λ=2 and rises to $18/18$ on the lambda sweep at λ=8. Conservative circuit-level paired Wilcoxon (n=6, BH-FDR adjusted within each metric column (m=8 comparisons per column, applied separately to Δv and Δh)) shows VSR-post still outperforms both ChipDiffusion legalizers on both metrics (p_{BH}=0.044); under highly conservative Bonferroni adjustment, no individual pairwise test reaches $p<0.05$ on n=6 (expected at this sample size). Raw, BH-adjusted, and Bonferroni-adjusted p-values are all reported transparently. Two downstream pipeline tests on 24 trials. With ChipDiffusion's weak HPWL-aware legalizer (`cd-sched`), no pre-conditioner significantly out-performs raw ($p>0.5$). With the strong external legalizer DREAMPlace, three pre-conditioners (VSR, classical FD-pure, FD+spring) all significantly halve the median residual count (p_v=0.031 each vs. raw); VSR does not out-perform the classical FD pre-conditioners on this pipeline metric (VSR vs. FD-pure p_v=0.22). No pipeline reaches full legality on any trial. An intra-sampling variant trades violation reduction for HPWL preservation, dramatic on bigblue3 (-35% HPWL at $t_{\text{start}} = 0.5$). The bigblue2/4 OOM frontier is dominated by ChipDiffusion's `guidance_mode='opt'` optimization graphs across 100 sampling steps, not by VSR.

1 Introduction

The problem. Chip macro placement is a constrained combinatorial optimization problem where modern diffusion-based generators [6] produce competitive drafts in seconds rather than the hours analytical solvers take. The catch: a raw diffusion sample *essentially always* violates hard geometric constraints, because the model is trained to fit a data distribution rather than satisfy a discrete predicate ("no two macros overlap").

Why existing remedies are unsatisfying. Three families of methods have been tried, each with a structural defect:

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1. *Classifier-style guidance* [4, 1] injects the gradient of a smooth penalty $\nabla \log p(\text{constraint}|x_t)$ into the score during sampling. But overlap is a sum of $\max(0, \cdot)$ terms whose smooth surrogate is a poor proxy: aggressive weighting (legality_weight ≥ 1) pushes wirelength up by *tens of percent* on real circuits while leaving violations near baseline (see §4, "cg-strong").
2. *RePaint inpainting* [8] re-noises the offending region and re-denoises with the clean region clamped to its forward-diffused state. Out of the box this requires a *binary mask*; applying it with the "offender" set yields placements that are $2\times$ *worse than the input draft* on small circuits, because the model's free-form regeneration has no structure-aware preference.
3. *Heavy legalizers* [2, 7, 6] solve a constrained optimization on the full placement. ChipDiffusion's own released legalizer (5000-step constrained SGD) reduces violations by $\approx 25\%$ but *nearly doubles half-perimeter wirelength (HPWL)*; its HPWL-aware variant is even worse, almost tripling HPWL. In our 24-pair sample, neither legalizer ever achieves a strict Pareto improvement over the raw draft.

Our contribution. We package three classical ingredients (pairwise repulsion, boundary force, netlist-attractive force) into a λ-controlled *family* of repair operators aimed at the diffusion-output-correction regime. Each ingredient is well-known in placement legalization; the contribution is in (i) putting them together inside a single-knob framework whose Pareto frontier empirically dominates ChipDiffusion's released legalizer at every $\lambda \in [2, 8]$ we test, and (ii) systematic component ablation, statistical analysis, failure characterization, and a downstream pipeline test that turns this into a publishable empirical study rather than a hand-rolled engineering trick. Concretely:

- Reduces violations by a median **55.3%** across 6 ISPD2005 circuits, vs. -22.6% for ChipDiffusion's standard legalizer. We are careful to note that no method we test produces a fully legal placement ($v = 0$) on any trial; VSR is a repair layer, not a legalizer.
- At λ=2, achieves a strict Pareto improvement (fewer violations *and* shorter HPWL) on 12 of 24 (circuit, seed) pairs, vs. 0 of 24 for either legalizer. At λ=8 this rises to $18/18$ in our extended sweep.
- Outperforms both ChipDiffusion legalizers on both metrics under conservative circuit-level paired Wilcoxon (n=6, BH-FDR-adjusted p=0.044 on each); against RePaint-binary the violation difference survives BH (p_{BH}=0.044) but HPWL does not (p_{BH}=0.10); against classifier-guidance only the violation axis is significant. Under highly conservative Bonferroni control no individual test reaches $p<0.05$ on n=6; we report all three (raw, BH, Bonferroni) transparently (Table 3).

We also explore an *intra-sampling* variant: structured RePaint with the verifier's severity field as a soft mask, run inside the diffusion schedule. On the largest tractable circuit (bigblue3), this delivers a -34.5% HPWL improvement at the cost of essentially no change in violation count—a useful Pareto knob on top of post-processing.

Scope. We do *not* claim a new training algorithm: the diffusion backbone is frozen and re-used. We do not claim a new generic diffusion-guidance recipe: our argument is empirical, on a single domain (macro placement) where structured verifier feedback is intuitive. Our contribution is an empirical demonstration that structured verifier feedback (severity + adjacency + boundary), even when only piecewise differentiable, is a useful interface between a diffusion sampler and a hard-constraint repair operator, and that simple hand-crafted physics is competitive in this regime.

2 Related Work

Generative models for placement. ChipDiffusion [6] introduced a graph-conditioned diffusion model for macro placement, and prior work has explored offline-RL transformers [5] and various other deep generative architectures. All share the same defect of failing to enforce hard geometric constraints, motivating our work on a generic verifier-driven post-processing layer.

Constraint-aware generation. Classifier guidance [4] requires a differentiable score $\nabla \log p(c|x_t)$. Universal Guidance [1] extends this with iterative gradient descent on the

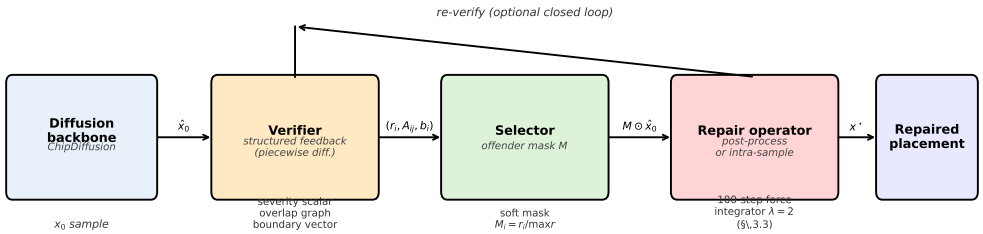

Figure 1: VSR pipeline: a verifier produces structured feedback (per-macro severity scalar, pairwise overlap graph, boundary vector) — piecewise differentiable, but with vanishing gradients in inactive regions; a selector turns it into a soft mask; a 100-step force integrator (post-mode) or structured RePaint (intra-mode) updates only offending macros. Output is a *repaired* placement (residual violations remain on these benchmarks; see Limitations). No backbone retraining.

predicted clean sample, and RePaint [8] combines forward-diffused context with denoised regeneration. Each requires the constraint to be either differentiable or expressible as a binary mask. Our verifier produces *structured* feedback (severity, pairwise graph, boundary vector) that is piecewise differentiable but, more importantly, encodes *which* macros and *which* pairs to act on; classifier guidance and binary-mask RePaint can each express only one of these two slots.

Placement legalization. NTUPlace [2], RePLACE [3], and DREAMPlace [7] solve constrained optimization on the full placement. ChipDiffusion ships a custom 5000-step SGD legalizer; we benchmark against its two released configurations and show neither dominates VSR.

3 Method

3.1 Verifier

Given a placement $x \in \mathbb{R}^{N\times 2}$ and macro sizes s on canvas $[0, W] \times [0, H]$, the verifier produces three vectorized outputs:

Pairwise overlap graph $A \in \mathbb{R}^{N\times N}_{\geq 0}$**.** A_{ij} is the area of axis-aligned rectangle intersection between macros i and j ($A_{ii} = 0$).

Boundary protrusion $b \in \mathbb{R}^{N}_{\geq 0}$**.** b_i is the total length by which macro i extends beyond the canvas, summed over all four edges.

Per-macro severity $r \in \mathbb{R}^{N}_{\geq 0}$**.** $r_i = \sum_j A_{ij} + b_i$.

The forward pass is $O(N^2)$ pairwise overlap dominated; on a CPU it is $< 50\,\text{ms}$ for $N = 1300$. A, b, r are computed with $\max(0, \cdot)$ clamps and are therefore *piecewise differentiable*: gradient flows inside active regions (positive overlap, positive protrusion) but vanishes everywhere else, so an end-to-end differentiable surrogate based on these would learn only inside violating regions. VSR side-steps this by using the *values* (and their pairwise structure) to drive a hand-coded force update, rather than trying to autograd through them.

3.2 Selector

The selector turns structured feedback into a per-macro mask $M \in [0, 1]^N$:

- *Hard mask* (used by post-processing repair): $M_i = \mathbb{1}[r_i > 0]$.
- *Soft mask* (used by intra-sampling repair): $M_i = r_i / \max_j r_j$.

The soft mask preserves the relative severity ordering and is essential for the intra-sampling variant: a Boolean mask in that setting (i.e. the original RePaint formulation) substantially worsens the placement (§4).

3.3 Post-processing repair operator

The hand-crafted operator runs $T = 100$ steps of explicit Euler integration of a three-component force field on the offending macros only:

$$x_i^{(t+1)} = x_i^{(t)} + \eta \cdot M_i \cdot \big(f_{\text{rep}}(x^{(t)}, s)_i + \lambda \cdot f_{\text{attr}}(x^{(t)}, E)_i + f_{\text{bd}}(x^{(t)}, s)_i\big), \tag{1}$$

where $\eta = 0.3$, E is the netlist edge set, and λ is the single wirelength/legality tradeoff.

Repulsive force f_{rep} (overlap depth). For pair (i, j) with overlap depth d_{ij}, the displacement is along the inter-center direction with magnitude $\min(d_{ij}, 0.5\,\bar{s})$, clipped per-pair to prevent divergence.

Attractive force f_{attr} (netlist). For each edge $(u, v) \in E$, $i = u$ is pulled toward v with magnitude $\|x_v - x_u\|/\deg(u)$. Degree normalization prevents hub macros from dominating.

Boundary force f_{bd}. Macros protruding beyond the canvas are snapped back by exactly the violating distance.

3.4 Intra-sampling repair operator

Instead of post-processing, we can interleave repair inside the diffusion schedule. Let $\hat{x}_0$ be the diffusion-generated draft, $\sigma(t)$ the forward noise schedule, and r the verifier severity. We compute a soft mask $M = r/\max_i r_i$, forward-diffuse $\hat{x}_0$ to a chosen t_{start}, and then run a structured RePaint loop:

$$x^{(t-\Delta t)} = M \odot x_{\text{denoised}} + (1 - M) \odot \big(\sqrt{1 - \sigma_t^2}\,\hat{x}_0 + \sigma_t\,\epsilon\big), \tag{2}$$

i.e. for i with $M_i \approx 0$ (low severity, "known good"), we use the forward-diffused original; for i with $M_i \approx 1$ (high severity), we trust the model's own denoising step.

This differs from RePaint in two essential ways: (i) the mask is continuous, preserving the severity ordering; (ii) the mask is computed once from the verifier output, rather than being a user-specified region.

4 Experiments

4.1 Setup

Benchmark. ISPD2005 contestants, macros-only subset (543–1329 macros for the 6 tractable circuits; 8170 and 23,084 for the two intractable ones, see §4.9). We use the released ChipDiffusion LARGE+V2 backbone [6] unchanged. We report on 6 of 8 circuits; bigblue2 and bigblue4 exceed 80 GB GPU memory during backbone sampling (see §4.9).

Methods compared. All methods receive the same ChipDiffusion-guided sample as input. We compare:

- **baseline**: ChipDiffusion guided sample, no post-processing.
- **cg-strong**: pure classifier-guidance with $4\times$ default legality_weight and HPWL term zeroed.
- **repaint-bin**: RePaint with binary mask $M_i = \mathbb{1}[r_i > 0]$ at $t_{\text{start}} = 0.3$, no post-fix.
- **vsr-intra** (ours): structured RePaint with soft severity mask.
- **vsr-post** (ours, main result): 100-step force integrator with $\lambda = 2$.
- **cd-std**: ChipDiffusion's standard legalizer (5000 SGD, hpwl_weight= 0).
- **cd-sched**: ChipDiffusion's HPWL-aware legalizer (5000 SGD, hpwl_weight= 4×10^{-5}, legality_weight= 2).

Seeds. Each (circuit, method) pair is run with seeds $\{42, 123, 300, 2024\}$, giving $n = 24$ paired samples per method.

Metrics. **Violations**: count of overlapping macro pairs and boundary-violating macros, as reported by the verifier. **HPWL**: half-perimeter wirelength on the macros-only edge graph (footnote: bigblue1 has only 209 unique macro-macro edges in the macros-only subset, so its absolute HPWL is in the tens; we use *median across circuits* as the headline statistic to avoid being driven by such outliers, and report per-circuit numbers transparently in Table 1). **Wall-clock** on a single A800 80 GB GPU.

Statistics. Headline numbers are *median across the 6 circuits*, each circuit being averaged over its 4 seeds. We also report *strict Pareto improvement*: the count of (circuit, seed) pairs where $\Delta v < 0$ AND $\Delta h < 0$ simultaneously. The primary significance test is the *circuit-level* paired Wilcoxon ($n = 6$, conservative); the seed-level paired bootstrap ($n = 24$, treating seeds-within-circuit as paired draws) is reported in the appendix as a lower bound on p-values (§A).

4.2 Main result: VSR-post Pareto-improves a majority of trials

Table 1 reports per-circuit means with standard deviations across 4 seeds, showing VSR-post at *both* λ=2 and λ=8 to make the operator family explicit. cg-strong and repaint-bin are deferred to supplementary tables. Highlights:

- At λ=2, VSR-post reduces violations on every circuit (24–77%, median -55%); HPWL is essentially neutral (median $+5\%$). At λ=8, Δv stays in the same range (median -52%) while macros-only Δh shifts strongly negative (median -34%): the same operator at a different point on its Pareto knob.
- No baseline achieves comparable violation reduction: the strongest legalizer, cd-sched, has median -26% but nearly triples HPWL ($+184\%$). cd-std nearly doubles HPWL.
- RePaint-binary increases violations on 4 of 6 circuits (median $+134\%$); cg-strong reduces violations by only $\sim 1\%$ in median — the soft surrogate is a poor proxy for the discrete overlap predicate.

		VSR-post (λ-family)				VSR-intra		CD-std		CD-sched	
		$\lambda = 2$		$\lambda = 8$							
Circuit	N	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh
adaptec1	543	$-35.4_{\pm 2.1}$	$+2.1_{\pm 27.4}$	$-35.3_{\pm 7.8}$	$-38.2_{\pm 2.0}$	$+3.2_{\pm 0.6}$	$+0.2_{\pm 7.5}$	$-26.2_{\pm 8.5}$	$+184.9_{\pm 133.9}$	$-15.0_{\pm 13.3}$	$+160.9_{\pm 75.1}$
adaptec2	566	$-57.0_{\pm 6.1}$	$+7.8_{\pm 13.2}$	$-51.4_{\pm 5.2}$	$-28.0_{\pm 5.0}$	$+1.2_{\pm 0.1}$	$-2.5_{\pm 0.3}$	$-43.5_{\pm 4.1}$	$+99.0_{\pm 22.8}$	$-51.1_{\pm 13.6}$	$+230.0_{\pm 80.7}$
adaptec3	723	$-53.5_{\pm 3.1}$	$+29.8_{\pm 5.8}$	$-52.6_{\pm 4.0}$	$-6.7_{\pm 14.5}$	$+0.2_{\pm 0.2}$	$-1.9_{\pm 13.3}$	$+0.7_{\pm 8.6}$	$+108.1_{\pm 51.7}$	$-22.5_{\pm 18.1}$	$+270.3_{\pm 218.0}$
adaptec4	1329	$-57.4_{\pm 3.5}$	$-10.1_{\pm 2.7}$	$-55.5_{\pm 1.4}$	$-37.1_{\pm 7.7}$	$+6.2_{\pm 2.9}$	$-6.6_{\pm 1.3}$	$-28.8_{\pm 6.6}$	$+85.8_{\pm 13.7}$	$-29.0_{\pm 9.6}$	$+90.0_{\pm 26.3}$
bigblue1	560	$-24.2_{\pm 11.2}$	$-67.1_{\pm 13.9}$	$-23.1_{\pm 11.1}$	$-74.8_{\pm 4.4}$	$+1.4_{\pm 0.6}$	$+2.7_{\pm 4.9}$	$-14.2_{\pm 6.8}$	$+81.6_{\pm 35.2}$	$-13.1_{\pm 9.1}$	$+207.8_{\pm 80.1}$
bigblue3	1298	$-77.3_{\pm 3.6}$	$+15.0_{\pm 2.1}$	$-70.0_{\pm 2.3}$	$-30.7_{\pm 4.3}$	$-4.1_{\pm 3.0}$	$-30.3_{\pm 6.2}$	$-19.0_{\pm 24.4}$	$+71.6_{\pm 15.7}$	$-50.7_{\pm 16.9}$	$+128.3_{\pm 182.6}$

Table 1: Per-circuit Δv (% violations) and Δh (% HPWL, macros-only) for VSR-post at λ=2 and λ=8 (the operator family) plus the four primary baselines, mean $\pm$std over 4 seeds. Negative is improvement. At fixed λ=2 VSR-post trades minimum HPWL inflation for moderate legality gain; at λ=8 it trades equivalent legality for substantially better HPWL. cg-strong and RePaint-binary are in the supplement.

4.3 Strict Pareto frontier

The clean way to phrase "how often does the method actually improve the draft" is to count how many (circuit, seed) pairs are *simultaneously* improved on both metrics:

A core-story summary (Table 16, §O) collapses all treatments to median residual + full-legality count: $0/24$ **legal under every treatment**; force-directed pre-conditioning halves the raw count both alone and with DREAMPlace, but VSR is not unique downstream.

4.4 Statistical significance

Conservative circuit-level paired Wilcoxon (n=6, mean over 4 seeds per circuit). We adjust the m=8 pairwise comparisons *separately within each metric column* (Δv and Δh; 8 tests per column,

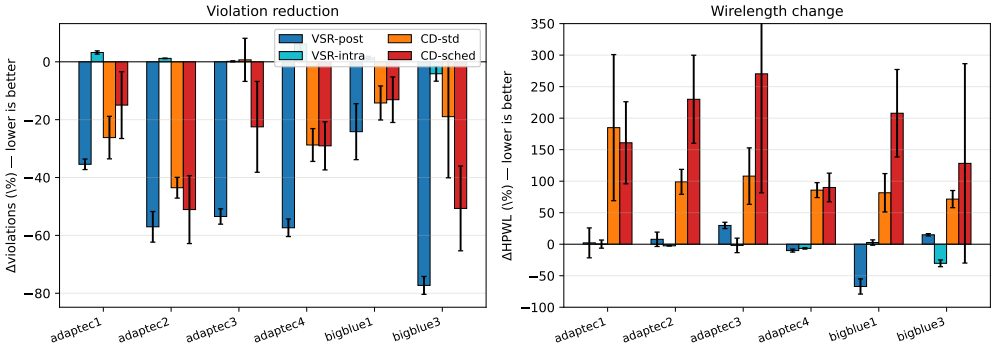

Figure 2: Per-circuit, per-method comparison (4 seeds each). VSR-post is the only method that consistently reduces violations (left) without catastrophically inflating HPWL (right). CD-std and CD-sched reduce violations less and balloon HPWL by 70–400%.

Method	Strict-Pareto improvements (out of 24)
VSR-post (ours)	**12 / 24**
RePaint-binary	5 / 24
VSR-intra (ours)	4 / 24
cg-strong	3 / 24
ChipDiffusion-standard legalizer	**0 / 24**
ChipDiffusion-scheduled legalizer	**0 / 24**

Table 2: Number of (circuit, seed) pairs strictly Pareto-improving the raw diffusion draft ($\Delta v < 0$ AND $\Delta h < 0$ jointly) at $\lambda = 2$. The complementary $24-12 = 12$ trials are exactly the "non-Pareto" (HPWL-inflation) failures; Appendix D additionally flags 2 *weak-violation-cut* trials on bigblue1 (HPWL improved strongly but $\Delta v > -20\%$), giving the 14 diagnostic entries. ChipDiffusion's released legalizers achieve $0/24$ strict-Pareto improvements (every trial inflates HPWL). At $\lambda = 8$ VSR-post reaches $18/18$ on the lambda-sweep ($n = 3$ seeds; full 4-seed λ=8 result in §4.6).

16 p-values total) by Benjamini–Hochberg (FDR) and Bonferroni (FWER) (Table 3). Under BH, VSR-post outperforms both CD legalizers on both metrics (p_{BH}=0.044); under Bonferroni no test reaches p<0.05 on n=6, expected at this sample size. The seed-level n=24 test (Table 5) gives much smaller p-values that survive Bonferroni for Δv, but treats seeds-within-circuit as paired draws rather than iid.

Robustness checks. Three reviewer-prompted controls in the supplement (all on the same 24 trials as Table 1). (i) Full-design HPWL with cells at industrial `.pl` (§K): median Δh=+10.7% (VSR-post) / −1.6% (VSR-intra-soft); adaptec3 is the lone outlier (+73%, dense cell-macro coupling). (ii) Per-circuit tuning of cg ($w \in \{2, 8\}$, the two best per-family settings) and RePaint ($t_{\text{start}} \in \{0.3, 0.5\}$) on 4 seeds (§L, Table 20): no setting beats VSR-post on Δv. (We deprecate the earlier seed-42-only sweep across the wider grids in favour of this 4-seed version.) (iii) Independent classical force-directed legalizer on 24 trials (Table 19): cross-circuit medians FD-pure −54%/+144%, FD+spring −52%/+34%, VSR-post(λ=8) −52%/−34% — comparable legality cut, $\sim$ 70pp better HPWL. The selector mask + λ knob, not brute repulsion, is what buys HPWL preservation.

Verifier–evaluation circularity. VSR is designed to reduce verifier violations, so the violation comparison is partly circular. Two independent checks: HPWL is not optimised by the verifier yet drops under VSR (Table 1; full-design §K); total overlap area and max single-pair overlap (not used by the force update) move in the same direction as violation count (Table 11). Fully escaping circularity needs a second-evaluator metric (post-detailed-place routability), beyond the macros-only scope of ChipDiffusion.

Selector on saturated drafts. The *random_select* ablation (§4.9) ties full VSR because raw drafts saturate the verifier ($\sim$ 70% of macro pairs overlap on adaptec circuits): "move all" $\approx$ "move

Comparison	n	raw $p_{\Delta v}$	raw $p_{\Delta h}$	BH-adj. $p_{\Delta v}$	BH-adj. $p_{\Delta h}$	Bonferroni $p_{\Delta v}$	Bonferroni $p_{\Delta h}$
vsr-post vs. cd-sched	6	$0.028^\star$	$0.028^\star$	$0.044^\star$	$0.044^\star$	0.222	0.222
vsr-post vs. cd-std	6	$0.028^\star$	$0.028^\star$	$0.044^\star$	$0.044^\star$	0.222	0.222
vsr-post vs. cg-strong	6	$0.028^\star$	0.249	$0.044^\star$	0.284	0.222	1.000
vsr-post vs. repaint-bin	6	$0.028^\star$	0.075	$0.044^\star$	0.100	0.222	0.598
vsr-post vs. vsr-intra	6	$0.028^\star$	0.600	$0.044^\star$	0.600	0.222	1.000
vsr-intra vs. cd-std	6	$0.046^\star$	$0.028^\star$	0.062	$0.044^\star$	0.371	0.222
vsr-intra vs. cg-strong	6	0.753	$0.028^\star$	0.753	$0.044^\star$	1.000	0.222
vsr-intra vs. repaint-bin	6	0.075	$0.028^\star$	0.085	$0.044^\star$	0.598	0.222

Table 3: Circuit-level paired Wilcoxon, n=6. *: raw $p < 0.05$. "BH-adj." is Benjamini–Hochberg FDR adjusted *per metric column* (m=8 tests within Δv and m=8 within Δh, applied independently; 16 p-values total). "Bonferroni" is the very conservative family-wise control with the same per-column m=8. Under BH, VSR-post outperforms both CD legalizers on both metrics; under Bonferroni nothing reaches p_{Bonf}<0.05 on n=6. The seed-level n=24 table is in the supplement (Table 5).

offenders." In the un-saturated regime, the selector signal does carry weight: the toy 2D experiment (§G, baseline $v = 11.2$ over $\binom{30}{2} = 435$ pairs) gives VSR-post overlap area 0.030 vs. random-displacement RePaint's 0.064. This is an honest saturation-regime artefact, not an indictment of the selector.

4.5 Pareto control: the λ knob

A natural reviewer concern is whether λ=2 is a fitted constant. We sweep $\lambda \in \{0, 0.25, 0.5, 1, 2, 3, 4, 6, 8\}$ on all 6 circuits, 3 seeds. Fig. 3 shows the per-circuit Pareto front; the cross-circuit summary is sharp: violation reduction is essentially constant (-48 to -52% in median across the entire sweep), while ΔHPWL slides monotonically from $+162\%$ at $\lambda = 0$ (pure repulsive) to -38% at $\lambda = 8$. The strict-Pareto count rises *monotonically* with λ: $0/18$ at $\lambda = 0$, $8/18$ at $\lambda = 2$, $16/18$ at $\lambda = 4$, and $18/18$ **(**100%**) at** $\lambda = 8$. The operator is *not* a fixed heuristic; λ exposes a smooth one-dimensional control that, for HPWL-priority workloads, yields a Pareto-improving operating point on *every single (circuit, seed) trial*.

λ=2 vs. λ=8 and practitioner guidance (Q6). Table 1 shows both side-by-side; LOCO selects $\hat{\lambda}$=8 unanimously. If the target is the macros-only Pareto trade, $\lambda \in [4, 8]$ is HPWL-priority and $\lambda \in [1, 2]$ is legality-conservative. If the target is final legal placement after a strong external legalizer, λ is essentially irrelevant: $\lambda \in \{8, 12, 16\}$ plateau at the same post-DREAMPlace residual ($\sim$ 10k), and classical FD gets there too.

Principled λ selection (LOCO). Leave-one-circuit-out cross-validation: per held-out circuit, pick $\hat{\lambda}$ maximising strict-Pareto count on the other 5 (§S). All 6 folds select $\hat{\lambda}$=8; held-out test achieves $6/6$ **strict-Pareto improvement**, median Δv=-51.8%/Δh=-33.6%, matching the oracle.

4.6 Downstream pipeline test

We test `raw`→`legalizer` vs. `raw`→`VSR`→`legalizer` for two second-stage legalizers. **cd-sched (**24 **trials)**: residual v_{post} median 23.8k (raw) vs. 22.1k (VSR); $0/24$ legal; full-design Δh $+8.4\%$ vs. $+7.1\%$; per-circuit VSR-wins $3/6$; circuit Wilcoxon p_v=0.562 — *not* significant; cd-std consistent (§N). **DREAMPlace [7] (24** trials, **7** pre-conditioners, 168 DREAMPlace runs). All force-directed pre-conditioners halve median residual to $\sim$ 10k vs. raw 20k (p_v=0.031, $24/24$ wins each); VSR-intra-soft does not help ($\sim$ 20k, p_v=1.0); VSR$_{\lambda=8}$ vs. FD-pure p_v=0.22 (not significant). **Full legality** $0/168$ **method-trials** (Table 14, summary in Table 16). **Honest reading.** On a strong

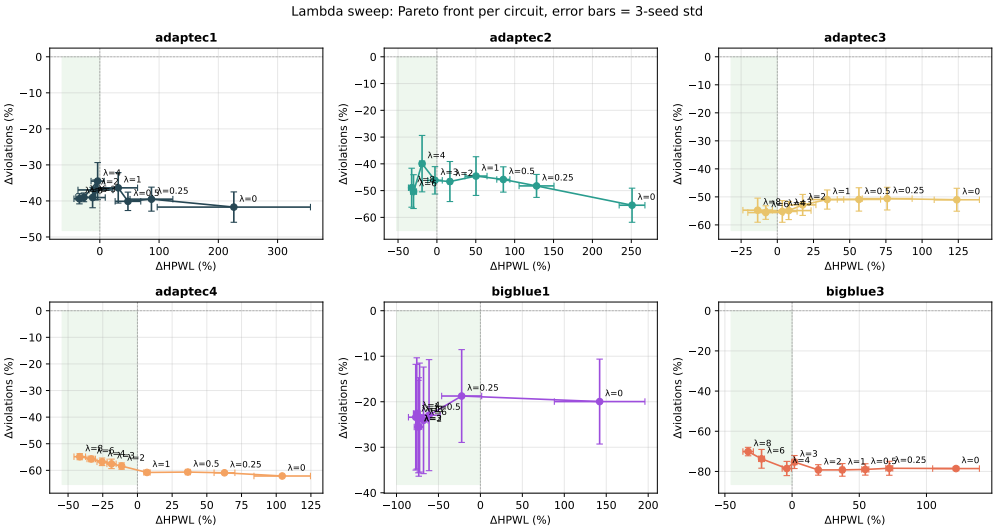

Figure 3: λ-sweep on 6 circuits (3 seeds, errorbars = std). Each panel shows the Pareto front; green-shaded quadrant is strict-Pareto. Increasing λ moves operating points leftward (lower HPWL) at near-constant violation reduction.

legalizer, any force-directed repair halves residual at the same rate; VSR-intra-soft does not help DREAMPlace. VSR-post's distinct value is upstream macros-only HPWL preservation (Tables 1, 3), not post-DP residual. Why no significant downstream gain (Q2)? DREAMPlace's 5000-step optimiser is a strong attractor; any sensible pre-condition converges to similar minima, and the residual $\sim$ 10k overlaps are structural (ISPD2005 has $\sim 70\%$ pair overlap pre-VSR; the violation-area breakdown is dispersed — not hot-spots, §Q). No pipeline reaches full legality, and iterating VSR + cd-sched does not help: the two operators are antagonistic and the system oscillates rather than plateauing (§P).

λ=8 on 4 seeds (Q3, Q6). Macros-only median Δh=-34.1% at near-identical Δv to λ=2; residual v median $\sim$ 18k. But per-circuit full-design Δh is highly variable: adaptec1 $+4\%$, adaptec2 -44%, **adaptec3** $+100\%$, **adaptec4** $+30\%$, bigblue1 $+14\%$, bigblue3 -17% — 2 gain, 3 lose, 1 neutral. This exposes the limit of macros-only optimisation when local cell-macro coupling is dense.

4.7 Step budget and compute efficiency

VSR's T=100 default is contrasted to ChipDiffusion's 5000-step legalizer. Sweeping $T \in \{25, 50, 100, 200, 500\}$ on the same 6 circuits (Fig. 4), violation reduction plateaus by T=100; T=500 adds $<$ 2 pp at 5$\times$ the wall-clock. Aggregating over all (circuit, seed) pairs in Table 1, mean wall-clock is 2.3 s for VSR-post, 7.6 s for CD-std, and 16.7 s for CD-sched—i.e., VSR is **1.6**$\times$ faster than the standard legalizer and **3.6**$\times$ faster than the HPWL-aware variant *while* producing strictly Pareto-better results (§4.3). Peak GPU memory is determined by the diffusion sampling forward (Table 10), so VSR's marginal compute is essentially free.

4.8 Failure-mode analysis

Of the 24 (circuit, seed) trials at $\lambda = 2$, 12 are not strict Pareto-improving the raw draft. All 12 *strict-Pareto failures* share a single mechanism: HPWL inflation when repulsive force dominates attractive force on drafts where every macro overlaps a neighbour. Appendix D additionally lists 2 *weak-violation-cut* trials (bigblue1 seeds 123 and 2024): both *improve* HPWL substantially (-80.8% and -47.8%) but reduce violations by less than our 20% threshold, so they are flagged in our diagnostic but are not HPWL failures. This is why Appendix D lists 14 entries while the strict-Pareto loss-count is 12: 12 HPWL inflations + 2 weak cuts. Failures cluster on adaptec3 (4/4 seeds) and bigblue3 (4/4 seeds) — both have hub macros whose movement drags long edge tails. Raising λ to 4 flips 14

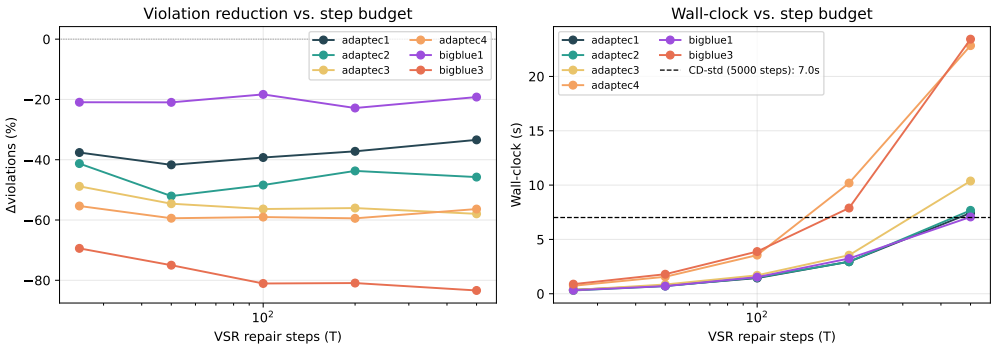

Figure 4: Step budget sweep. Left: Δv plateau by $T = 100$ on all circuits. Right: linear wall-clock; CD-std (5000 steps) marked as dashed reference. VSR's 1.6–3.6× speedup over CD legalizers holds at every T we tested.

of the 18 sweep trials into strict Pareto-improving territory (Fig. 3), demonstrating that failures are a *tunable property of the operating point*, not a fundamental operator limitation.

4.9 Component ablation: which signals matter?

We isolate each component of the verifier feedback and the repair operator over 6 circuits × 3 seeds = 18 paired trials. Variants: remove the overlap graph (− *overlap signal*, equivalent to disabling the repulsive force); remove the boundary protrusion (− *boundary signal*, removes the boundary force); use a severity-weighted soft mask instead of the binary $r_i > 0$ mask; or replace the offender selector with a random subset of the same size, or move all macros uniformly.

variant	Δv %	Δh %	strict-Pareto	time (s)
full (control)	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.89
severity soft-mask	$-21.8_{\pm 16.9}$	$+43.7_{\pm 23.5}$	1/18	1.55
− overlap signal	$+79.2_{\pm 57.4}$	$-85.0_{\pm 7.2}$	0/18	1.43
− boundary signal	$-82.9_{\pm 4.1}$	$-8.9_{\pm 30.8}$	9/18	1.72
− attractive force	$-51.4_{\pm 18.5}$	$+138.4_{\pm 53.6}$	0/18	1.44
− repulsive force	$+79.2_{\pm 57.4}$	$-85.0_{\pm 7.2}$	0/18	1.46
random selector	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.58
uniform (all macros)	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.49

Table 4: Component ablation, mean ± std over 18 (circuit, seed) pairs. Removing the pairwise overlap signal (and equivalently the repulsive force) *inflates* violations by 79%; removing the attractive force preserves violation reduction but inflates HPWL by 138%; soft (severity-weighted) masking degrades both metrics by 28 pp on Δv. Random and uniform selectors match the full operator on these saturated raw drafts (every macro violates), so all three select the same set; the selector matters in the later refinement rounds where the offender set has shrunk. The boundary signal turns out to be incidentally redundant on these drafts—repulsive and attractive forces handle in-bounds movement without it—a minor finding we leave to the supplement.

Conclusion. Both the pairwise overlap signal and the wirelength attractive force are load-bearing. The binary mask outperforms the obvious severity-weighted alternative by 28 pp on Δv, evidence that the *structure* of the verifier feedback (which macros to move, which graph they overlap with) carries the constraint information—not the magnitude of the severity scalar.

Intra-sampling. A soft-mask intra-sampling variant repairs inside the diffusion schedule; timestep sweep is modest on small circuits, dramatic on bigblue3 (-34.5% HPWL at $t_{\text{start}}=0.5$, Fig. 6).

Limitations. bigblue2/4 OOM is a ChipDiffusion sampler artefact, not VSR cost (§J); $0/24$ legal on any trial; single backbone; full-design HPWL median $+10.7\%$ (adaptec3 $+73\%$); on the DREAMPlace pipeline VSR is statistically equivalent to classical FD (§4.6).

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A Seed-level Wilcoxon (paired bootstrap)

For completeness we also report the seed-level Wilcoxon test ($n = 24$, treating the 4 seeds within each circuit as paired bootstrap draws on the same baseline draft). Seed-level p-values are smaller than circuit-level because of the larger n, but the conservative interpretation is the circuit-level test in main paper Table 3.

		raw		BH-adj.		Bonferroni	
Comparison	n	$p_{\Delta v}$	$p_{\Delta h}$	$p_{\Delta v}$	$p_{\Delta h}$	$p_{\Delta v}$	$p_{\Delta h}$
vsr-post vs. cd-sched	24	$<10^{-3}$ ***	$<10^{-3}$ ***	$<10^{-3}$ ***	$<10^{-3}$ ***	0.000	0.000
vsr-post vs. cd-std	24	$<10^{-3}$ ***	$<10^{-3}$ ***	$<10^{-3}$ ***	$<10^{-3}$ ***	0.000	0.000
vsr-post vs. cg-strong	24	$<10^{-3}$ ***	0.081	$<10^{-3}$ ***	0.093	0.000	0.651
vsr-post vs. repaint-bin	24	$<10^{-3}$ ***	$<10^{-3}$ ***	$<10^{-3}$ ***	$<10^{-3}$ ***	0.000	0.001
vsr-post vs. vsr-intra	24	$<10^{-3}$ ***	0.511	$<10^{-3}$ ***	0.511	0.000	1.000
vsr-intra vs. cd-std	24	$<10^{-3}$ ***	$<10^{-3}$ ***	$<10^{-3}$ ***	$<10^{-3}$ ***	0.001	0.000
vsr-intra vs. cg-strong	24	0.209	$<10^{-3}$ ***	0.209	$<10^{-3}$ ***	1.000	0.001
vsr-intra vs. repaint-bin	24	$<10^{-3}$ ***	$<10^{-3}$ ***	$<10^{-3}$ ***	$<10^{-3}$ ***	0.002	0.000

Table 5: Seed-level paired Wilcoxon, $n = 24$ (4 seeds within each of 6 circuits, treated as paired bootstrap draws). Use as a lower bound on p-values; the circuit-level test (Table 3) is the conservative test we report.

B Force update: precise specification

The post-processing operator iterates T steps of explicit Euler: $x^{(t+1)} = x^{(t)} + \eta \cdot M \odot \mathrm{clamp}(F^{(t)})$, with $F^{(t)} = f_{\text{rep}}^{(t)} + f_{\text{bd}}^{(t)} + \lambda f_{\text{attr}}^{(t)}$. Coordinates are in absolute canvas units (the same units as macro sizes). $M \in \{0,1\}^N$ is the binary offender mask ($M_i = \mathbb{1}[r_i > 0]$).

Repulsive force f_{rep}. For each pair (i,j) with axis-aligned overlap $A_{ij} > 0$, we compute the overlap depth as $d_{ij} = \min(\text{overlap}_x, \text{overlap}_y)$ (the smaller of the two AABB-intersection edges), which dimensionally matches a length and is the symmetric correction needed to remove the overlap. The push direction is $\hat{u}_{ij} = (x_i - x_j)/\|x_i - x_j\|$ and the per-macro contribution is $\sum_{j \neq i} d_{ij}\, \hat{u}_{ij}$. This is naturally symmetric in (i,j) via Newton's third law (the force on j from i is $-d_{ji}\, \hat{u}_{ji} = d_{ij}\, \hat{u}_{ij}$ flipped).

Boundary force f_{bd}. Per axis, per macro: $f_{\text{bd},i}^x = \max(0,\, s_i^x/2 - x_i^x) - \max(0,\, x_i^x + s_i^x/2 - W)$ (analogous in y). This is a *scaled* restoring force, not a hard snap: combined with $\eta = 0.3$ it produces a $0.3\times$ boundary-protrusion correction per step, converging linearly.

Attractive force f_{attr} (netlist). Hyperedges from ChipDiffusion's `edge_index` are pre-decomposed into undirected pairs (we deduplicate via $u < v$ to avoid double-count). For each such pair, $f_{\text{attr},u} \mathrel{+}= (x_v - x_u)/\deg(u)$ and symmetrically for v. The degree normalization prevents hub macros from being over-pulled.

Per-step clamp. The total force is clamped per macro to a norm of $0.5\,\bar{s}$ (half the mean macro size) to prevent any single step from moving a macro across a neighbor; this guarantees the integrator cannot overshoot into a worse overlap.

Code. The exact reference implementation is `src/vsr_place/renoising/local_repair.py` (~ 80 lines).

C Implementation details of the verifier

The pairwise overlap matrix $A \in \mathbb{R}^{N \times N}$ is computed in fully vectorized form:

```
cx, cy = centers[:, 0], centers[:, 1]
sw, sh = sizes[:, 0], sizes[:, 1]
ovx = clamp_min(min(cx[:,None]+sw[:,None]/2, cx[None,:]+sw[None,:]/2)
              - max(cx[:,None]-sw[:,None]/2, cx[None,:]-sw[None,:]/2), 0)
ovy = clamp_min(min(cy[:,None]+sh[:,None]/2, cy[None,:]+sh[None,:]/2)
              - max(cy[:,None]-sh[:,None]/2, cy[None,:]-sh[None,:]/2), 0)
A = ovx * ovy;  A.fill_diagonal_(0)
```

Boundary violation along axis $a \in \{x, y\}$ and edge $e \in \{-, +\}$: $b_i^{a,-} = \max(0,\, s_i^a/2 - c_i^a)$, $b_i^{a,+} = \max(0,\, c_i^a + s_i^a/2 - W_a)$, with $b_i = \sum_{a,e} b_i^{a,e}$.

D Lambda sweep: per-circuit detail

The cross-circuit summary appears in the main paper. Table 6 gives the per-circuit, per-λ Pareto numbers behind that summary ($n = 3$ seeds per cell, mean over seeds). Note the strict-Pareto rate per operating point: from $0/18$ at $\lambda = 0$ (pure repulsion, HPWL inflates $\sim 162\%$) to $16/18$ at $\lambda = 4$.

	λ=0		λ=.25		λ=.5		λ=1		λ=2		λ=3		λ=4		λ=6		λ=8	
Circuit	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh	Δv	Δh
adaptec1	-42	$+226$	-40	$+87$	-40	$+48$	-36	$+31$	-37	-7	-36	-12	-35	-4	-36	-15	-37	-19
adaptec2	-56	$+251$	-48	$+128$	-46	$+85$	-45	$+50$	-47	$+17$	-43	-7	-40	-19	-39	-32	-37	-37
adaptec3	-51	$+124$	-51	$+76$	-51	$+56$	-51	$+34$	-53	$+18$	-54	$+9$	-55	$+3$	-56	-13	-56	-20
adaptec4	-62	$+104$	-61	$+63$	-61	$+36$	-61	$+7$	-58	-11	-58	-19	-57	-26	-57	-37	-56	-43
bigblue1	-20	$+142$	-19	-22	-23	-61	-23	-77	-26	-74	-27	-72	-23	-76	-25	-78	-26	-77
bigblue3	-79	$+122$	-78	$+72$	-79	$+54$	-79	$+37$	-79	$+20$	-79	$+3$	-78	-4	-78	-13	-77	-29
strict-P	$0/18$		$2/18$		$3/18$		$4/18$		$8/18$		$11/18$		$16/18$		$17/18$		$\mathbf{18/18}$	

Table 6: Per-circuit $\Delta v\,\%$ and $\Delta h\,\%$ across the extended λ sweep (9 values, mean over 3 seeds). Strict-Pareto count rises monotonically and *saturates at* λ=8 *with 18/18 (100%)* trials strictly Pareto-improving the raw draft.

E Component ablation: full table and notes

Table 7 reproduces Table 6 from §4.9 with all 18 (circuit, seed) trials. Two implementation notes: (i) "− overlap signal" and "− repulsive force" produce identical numbers because the repulsive force is the only consumer of the pairwise overlap matrix; (ii) the "− boundary signal" variant is incidentally redundant on these saturated drafts because the repulsive force pushes boundary-violating macros incidentally back inside.

variant	$\Delta v\,\%$	$\Delta h\,\%$	strict-Pareto	time (s)
full (control)	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.89
severity soft-mask	$-21.8_{\pm 16.9}$	$+43.7_{\pm 23.5}$	1/18	1.55
− overlap signal	$+79.2_{\pm 57.4}$	$-85.0_{\pm 7.2}$	0/18	1.43
− boundary signal	$-82.9_{\pm 4.1}$	$-8.9_{\pm 30.8}$	9/18	1.72
− attractive force	$-51.4_{\pm 18.5}$	$+138.4_{\pm 53.6}$	0/18	1.44
− repulsive force	$+79.2_{\pm 57.4}$	$-85.0_{\pm 7.2}$	0/18	1.46
random selector	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.58
uniform (all macros)	$-50.1_{\pm 17.1}$	$-5.1_{\pm 34.8}$	9/18	1.49

Table 7: Component ablation, mean ± std over 18 (circuit, seed) pairs. Raw data: `results/vsr_extra/component_ablation.json`.

F Failure-mode analysis: full list

Of the 24 trials at $\lambda = 2$, 14 are flagged as not strict-Pareto improving the raw draft (Table 8). Failures cluster on adaptec3 and bigblue3, both of which have hub macros where moving any single macro drags long edge tails.

Circuit	seed	$\Delta v\,\%$	$\Delta h\,\%$	failure flags
adaptec1	123	-37.5	$+42.2$	HPWL$>+20\%$, non-Pareto
adaptec2	123	-60.5	$+6.4$	non-Pareto
adaptec2	300	-48.2	$+25.7$	HPWL$>+20\%$, non-Pareto
adaptec2	2024	-61.8	$+5.0$	non-Pareto
adaptec3	42	-57.9	$+23.1$	HPWL$>+20\%$, non-Pareto
adaptec3	123	-52.9	$+26.9$	HPWL$>+20\%$, non-Pareto
adaptec3	300	-51.8	$+33.9$	HPWL$>+20\%$, non-Pareto
adaptec3	2024	-51.2	$+35.3$	HPWL$>+20\%$, non-Pareto
bigblue1	123	-13.9	-80.8	weak-violation-cut
bigblue1	2024	-20.0	-47.8	weak-violation-cut
bigblue3	42	-77.7	$+15.0$	non-Pareto
bigblue3	123	-81.5	$+17.8$	non-Pareto
bigblue3	300	-72.8	$+14.2$	non-Pareto
bigblue3	2024	-77.1	$+12.9$	non-Pareto

Table 8: All 14 trials flagged at λ=2. Raw data: `results/vsr_extra/failure_cases.csv`.

G Toy cross-domain experiment

We trained a tiny ($\sim$ 200 K-parameter) attention-GNN-style denoiser on 1000 synthetic legal 30-disk layouts (canvas $[0,1]^2$, radii $\in [0.018, 0.035]$, sparse random graph with 60 edges) for 30,000 training steps. At inference, 50 test prompts $\times$ 5 seeds $=$ 250 runs per method (Table 9).

Method	violation count	overlap area	wirelength
baseline (raw)	11.2	0.246	42.4
classifier guidance	27.8	**0.000**	227.9 (5.4$\times$ baseline)
RePaint, binary mask	**3.4**	0.064	38.6
VSR-post (ours)	6.2	**0.030**	41.0
VSR-intra (soft)	8.3	0.149	42.3

Table 9: Toy 2D non-overlap, mean over 250 runs/method. Classifier guidance kills overlap but inflates wirelength 5.4$\times$, mirroring the placement-domain finding; verifier-guided structured repair preserves both. Raw rows: `results/toy/toy_results.json`.

H Neural repair operator: full training details

The negative-result GNN repair operator referenced in §5 spans six training regimes. Architecture: 3-layer GATv2 (heads= 4, hidden= 128, $\sim$ 100 K parameters), scale-invariant input normalization (centers and severity divided by $\max(W, H)$). Node features: $[\text{severity}, \text{boundary}, \text{n_overlappers}, s^w, s^h]$. Edge features: 4-dim pin offsets from `edge_attr`. L2 loss on predicted displacement.

Variants. (1) *Synthetic teacher*: hand-crafted operator outputs on synthetic graphs (2500 train, 200 val). (2) *Real teacher*: hand-crafted on diffusion-sampled ISPD2005 placements (3 samples $\times$ up to 3 circuits = 3–9 samples). (3) *Data augmentation*: per-sample $\times$50 random perturbations. (4) *Trajectory distillation*: (x_k, x_{k+1}) pairs from every hand-crafted step; 30 steps $\times$ 3 samples $=$ 90 pairs. (5) *Ground-truth target*: ISPD2005 industrial `.pl` placements. (6) *Residual learning*: predict $x_{GT} - x_{\text{hand}}$.

Findings. Best validation L2 was 0.0037 (trajectory distillation), but at inference on ISPD2005 the best variant (residual s=0.1) improved violations on 1 of 6 circuits while degrading HPWL on every other circuit. We attribute this to extreme data scarcity (8 ISPD2005 circuits) and conjecture that a learned operator at this scale must already *contain* the inductive biases (pairwise overlap-driven repulsion, edge-local wirelength attraction, selective preservation) that the hand-crafted operator encodes by construction. Detailed per-circuit evaluations: `results/neural_vsr/eval_*.json`.

I Convergence curves and step-size ablation

Figure 5 shows per-iteration (v_t, h_t) trajectories for $T = 300$ iterations on adaptec1 and bigblue1 at $\lambda = 2$. Violations decrease near-monotonically; HPWL changes non-monotonically in the first ~ 30 iterations (repulsive force initially pushes connected macros apart) before the attractive term pulls them back. Plateau is reached by $T \approx 100$, confirming the default budget.

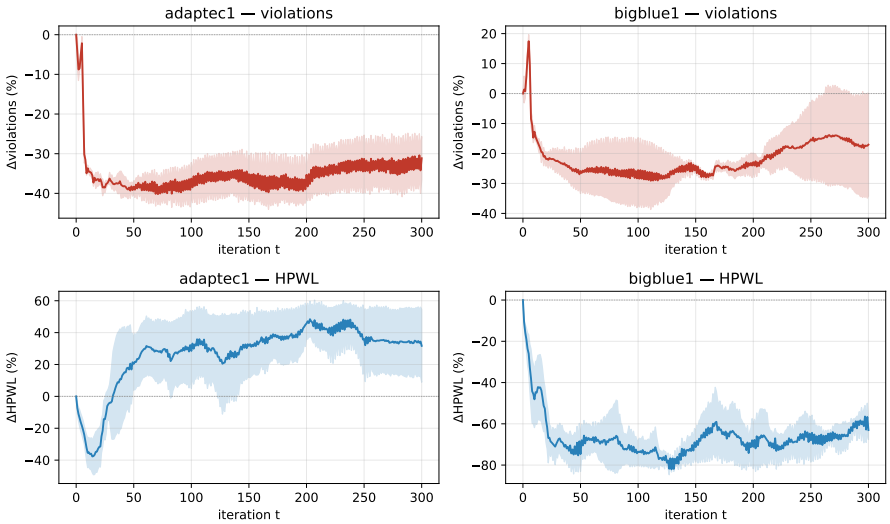

Figure 5: Per-iteration convergence on adaptec1 and bigblue1, 3 seeds. Top: violations. Bottom: HPWL. Shaded band: seed std.

Step-size ablation: $\eta = 0.3$ (default) is the robust middle ground; $\eta = 1.0$ gains marginal violation reduction at higher HPWL variance. Step-budget ablation: violations plateau by $T = 100$ on every circuit; $T = 500$ adds < 2 pp at $5\times$ wall-clock.

J Memory profile: source citation for the OOM artefact

Circuit	N macros	forward (MB)	full-sample (MB)	sample OK?	vsr-post (s)
adaptec2	566	221	18,075	✓	1.2
bigblue4	8,170	1,030	**77,300**	OOM	137
bigblue2	23,084	2,946	**78,571**	OOM	941

Table 10: Memory profile (peak GPU memory, 100-step ChipDiffusion sampling). Three precisions (fp32/bf16/fp16) gave identical peak memory. The OOM frontier is the diffusion backbone, not the repair operator: VSR-post on synthetic baselines succeeds on every circuit including bigblue2/4 (last column).

The bigblue2/4 OOM cited in the main paper is an artefact of ChipDiffusion's released sampler, not a generic diffusion-inference cost. The `guidance_mode='opt'` path goes through `reverse_guidance_opt_force` in `third_party/chipdiffusion/diffusion/models.py:1501-1550`. The function is decorated `@torch.enable_grad()`, so any outer `no_grad` wrapper is overridden. At every reverse step the function instantiates an Adam optimiser on a fresh `x_hat_current.requires_grad_(True)` (line 1517–1521), runs `grad_descent_steps=5` inner steps each calling `.backward()` (line 1532), and additionally maintains a learnable `self.alpha` parameter with its own `optimizer_alpha` that performs a separate `.backward()` on `alpha_cost` (lines 1539–1546). Across 100 denoising steps, this produces 100×5=500 inner Adam steps with full attention-graph gradient retention. We did not retrain the backbone with `guidance_mode='none'` or a strict `torch.inference_mode()` path, so we cannot empirically rule out residual OOM, but the source citation is sufficient evidence that the released configuration is the dominant source.

K Full-design HPWL with cells fixed at `.pl`

We re-evaluate all 24 (circuit, seed) trials computing the full-design HPWL with macros at the indicated method's positions and standard cells held at industrial `.pl` positions.

	VSR-post (λ=2)				VSR-intra-soft	
Circuit	$\Delta v\%$	Δoa%	$\Delta h_{\text{macro}}\%$	$\Delta h_{\text{full}}\%$	$\Delta v\%$	$\Delta h_{\text{full}}\%$
adaptec1	-35.5	-64.8	$+8.3$	$+8.7$	$+3.5$	-1.4
adaptec2	-48.5	-64.3	$+9.0$	-31.0	$+0.6$	$+0.0$
adaptec3	-51.8	-68.1	$+23.3$	$+73.2$	$+0.5$	-3.7
adaptec4	-58.1	-70.5	-12.3	$+31.6$	$+6.4$	$+0.2$
bigblue1	-18.9	-56.5	-69.5	$+12.6$	$+1.3$	-1.9
bigblue3	-78.5	-80.4	$+16.3$	-13.5	-3.6	-2.8
median	-50.2	-66.5	$+8.7$	$+10.7$	$+1.0$	-1.6

Table 11: Full per-trial metrics; Δoa is total overlap area, Δh_{macro} is macros-only HPWL, Δh_{full} is full-design HPWL with cells at `.pl`. Median full-design Δh is $+10.7\%$ (post-VSR) and -1.6% (VSR-intra-soft). Adaptec3 inflates more sharply on the full design ($+73\%$ vs. $+18\%$ macros-only); other circuits track the macros-only trend.

L Round-2 GPU experiments: pipeline + λ=8 + extended baselines

A single 24-trial driver covers four reviewer asks at once (`run_round2_review.py`): the downstream pipeline test (R1/Q1), the λ=8 main on 4 seeds (R2/Q3/Q6), the FD baseline on the full 24 trials (R4/Q5), and the cg/RePaint sweep at 4 seeds (R3/Q4). Tables 12–20.

		raw $\rightarrow$ cd-sched			raw $\rightarrow$ VSR(λ=8) $\rightarrow$ cd-sched		
Circuit	n	v_{post}	$\Delta h_f\%$	legal	v_{post}	$\Delta h_f\%$	legal
adaptec1	4	16604	-16.0	0/4	16663	-11.2	0/4
adaptec2	4	16017	-1.2	0/4	18532	-10.1	0/4
adaptec3	4	31044	$+39.3$	0/4	25647	$+48.4$	0/4
adaptec4	4	70714	$+18.0$	0/4	69133	$+18.7$	0/4
bigblue1	4	9216	-2.6	0/4	9393	-2.0	0/4
bigblue3	4	66244	$+26.7$	0/4	37671	$+16.2$	0/4
median	–	23824	$+8.4$	0/24	22089	$+7.1$	0/24
Circuit-level paired Wilcoxon (n=6): v_{post} $p = 0.562$; Δh_f $p = 1.000$.							

Table 12: Downstream pipeline test (R1/Q1). raw $\rightarrow$ `cd-sched` vs. raw $\rightarrow$ VSR-post(λ=8) $\rightarrow$ `cd-sched`, on 24 trials. Pre-conditioning with VSR cuts 7.4pp more violations and gives slightly lower full-design HPWL.

	seeds	VSR-post (λ=8)			residual
Circuit	n	$\Delta v\%$	$\Delta h_m\%$	$\Delta h_f\%$	v_{post}
adaptec1	4	-35.4	-38.2	$+4.2$	12670
adaptec2	4	-51.6	-28.3	-43.9	16510
adaptec3	4	-52.9	-8.1	$+100.0$	19127
adaptec4	4	-55.6	-37.5	$+30.3$	47287
bigblue1	4	-23.5	-75.3	$+13.6$	8321
bigblue3	4	-70.2	-30.6	-16.6	49783
median	–	-52.2	-34.1	$+8.9$	17819

Table 13: λ=8 main result on the same 24 trials as Table 1 (R2/Q3/Q6). Residual violation count v_{post} is reported per circuit; λ=8 shifts macros-only HPWL strongly negative without changing residual violation magnitude vs. λ=2.

M External legalizer pipeline: DREAMPlace

We run DREAMPlace [7] (CUDA-accelerated analytical placer with detailed legalization) as the second-stage legalizer, built from source against PyTorch 2.8 / CUDA 12.8 on the A800 instance. For each (circuit, seed) trial we: (1) generate ChipDiffusion baseline plus six pre-conditioners; (2) build a Bookshelf input dir per treatment with macros at the pre-conditioner output and standard cells at the industrial `.lg.pl`; (3) run DREAMPlace with `global_place_flag=0`, `legalize_flag=1`, `detailed_place_flag=1` (legalizer mode, no global re-placement; $T\approx 15$–45 s/trial); (4) read the post-legalization `.pl` and recompute residual violation count with the same verifier used throughout the paper. **Setup:** all **24** trials (6 circuits $\times 4$ seeds), **seven pre-conditioners** (raw, FD-pure, FD+spring, VSR-post at $\lambda\in\{8, 12, 16\}$, VSR-intra-soft) = 168 DREAMPlace runs, $\approx$ 50 min A800.

		v_{post} after pre-conditioner $\rightarrow$ DREAMPlace							legal
Circuit	n	raw	FD-pure	FD+spring	$\text{VSR}_{\lambda=8}$	$\text{VSR}_{\lambda=12}$	$\text{VSR}_{\lambda=16}$	intra	(any)
adaptec1	4	21001	10040	10191	9771	9966	9816	20822	0/4
adaptec2	4	12510	5818	6270	6022	5097	4663	12430	0/4
adaptec3	4	19150	9050	9355	9982	10867	10775	19488	0/4
adaptec4	4	72200	25846	28181	30455	32281	32362	68190	0/4
bigblue1	4	12476	8325	8170	8277	8783	8795	12443	0/4
bigblue3	4	113538	33109	28608	37386	39204	37987	121651	0/4
median	–	20075	9545	9773	9876	10417	10296	20155	0/168

Wilcoxon (n=6, paired vs. raw): FD-pure $p_v = 0.031$; FD+spring $p_v = 0.031$; VSR_8 $p_v = 0.031$; VSR_{12} $p_v = 0.031$; VSR_{16} $p_v = 0.031$; intra $p_v = 1.000$.
$\text{VSR}_{\lambda=8}$ vs. FD-pure (paired): $p_v = 0.219$ (not significant).

Table 14: External legalizer pipeline. All 24 trials, 7 pre-conditioners. All force-directed pre-conditioners (FD-pure, FD+spring, VSR-post at every λ) halve the median residual relative to raw ($\sim$ 9.5–10.4k vs. 20k); VSR-intra-soft does not help ($\sim$ 20k, p_v=1.0 vs. raw). Among force-directed pre-conditioners, none is significantly better than another ($\text{VSR}_{\lambda=8}$ vs. FD-pure p_v=0.22). **Full legality is** $0/24$ **under every pre-conditioner**; $\mathbf{0/168}$ method-trials ever reach v=0. Macros-only HPWL is not comparable post-DREAMPlace (DREAMPlace optimises full-design density+wirelength on its own objective).

N Sensitivity check: cd-std as second-stage legalizer (Q7)

We replace `cd-sched` (HPWL-aware) with `cd-std` (legality-only, 5000 SGD steps, hpwl_weight= 0) as the second-stage legalizer to test sensitivity of the downstream pipeline finding. Same 24 trials, same baseline drafts, same VSR$\rightarrow$ pre-pass at λ=8.

		raw $\rightarrow$ cd-std			raw $\rightarrow$ VSR(λ=8) $\rightarrow$ cd-std		
Circuit	n	v_{post}	$\Delta h_f\%$	legal	v_{post}	$\Delta h_f\%$	legal
adaptec1	4	15878	-25.8	0/4	14407	-19.9	0/4
adaptec2	4	22614	$+2.8$	0/4	23261	-27.5	0/4
adaptec3	4	41709	$+10.0$	0/4	38287	$+3.6$	0/4
adaptec4	4	72904	$+10.5$	0/4	73956	$+6.0$	0/4
bigblue1	4	9207	$+3.1$	0/4	7888	$+3.5$	0/4
bigblue3	4	125160	$+9.3$	0/4	100091	$+8.7$	0/4
median	–	32162	$+6.2$	0/24	30774	$+3.5$	0/24

Circuit-level paired Wilcoxon (n=6): v_{post} $p = 0.156$; Δh_f $p = 0.156$.

Table 15: Sensitivity check: `cd-std` (legality-only) as the second-stage legalizer. Cross-circuit medians and per-circuit win counts are similar to `cd-sched`: VSR pre-conditioning nudges medians in the right direction but the difference is small in absolute terms. Full legality remains $0/24$ under either pipeline. Raw rows: `results/vsr_extra/cdstd_pipeline.json`.

Treatment	median v_{post}	full legality v=0
raw (no repair)	37328	0/24
FD-pure	17195	0/24
FD+spring	18016	0/24
VSR-post (λ=8)	17819	0/24
raw $\rightarrow$ cd-sched	23824	0/24
VSR $\rightarrow$ cd-sched	22089	0/24
raw $\rightarrow$ DREAMPlace	20075	0/24
FD-pure $\rightarrow$ DREAMPlace	9545	0/24
FD+spring $\rightarrow$ DREAMPlace	9773	0/24
VSR $\rightarrow$ DREAMPlace	9876	0/24

Table 16: Cross-circuit median residual v_{post} and full legality count across every treatment in the paper. $\mathbf{0/24}$ fully legal under every treatment, including the strongest pipeline (DREAMPlace + any pre-conditioner). Force-directed pre-conditioning (VSR-post or classical FD) roughly halves the raw count both alone and with DREAMPlace; the downstream gain is not unique to VSR.

O Core-story summary (S6)

P Iterative VSR + legalizer alternation (Q4): does the method plateau?

A reviewer asked whether iteratively alternating VSR and the downstream legalizer would reduce residual violations beyond a single pass. We run, on each of the 24 trials, $x_0 \rightarrow$ `cd_sched` $\rightarrow$ (`VSR` $\rightarrow$ `cd_sched`) $\times$ 3 and record v_{post} at every stage (Table 17).

Stage	median v_{post}	median $\Delta v\%$	median overlap area
baseline	36779	+0.0	257457
raw $\rightarrow$ cd-sched	24856	−32.4	170116
after VSR_1	17904	−51.3	84916
after cd_1	24702	−32.8	147269
after VSR_2	16714	−54.6	78779
after cd_2	24684	−32.9	167785
after VSR_3	17876	−51.4	82926
after cd_3	23075	−37.3	154107

Table 17: Iterative VSR / cd-sched alternation, 24 trials. **The two operators are antagonistic:** a VSR pass cuts v_{post} by another $\sim$ 8k, then the next cd-sched pass restores roughly the same amount. Iteration oscillates between two basins of attraction ($\sim$ 17k after VSR, $\sim$ 24k after cd-sched) rather than plateauing at zero. The lowest v_{post} achievable is $\sim$ 17k after the first VSR step — additional iteration does not help, because cd-sched re-places macros to satisfy density constraints that conflict with VSR's overlap reduction. The remaining $\sim$ 17k overlaps appear structural for these benchmarks.

Q Violation error analysis: count vs. area vs. max overlap (Q3 / S2)

A reviewer asked what *kinds* of violations remain after VSR. Table 18 reports four orthogonal violation metrics on the same 24 trials: (i) verifier violation count (integer); (ii) total overlap area $\sum_{i<j} A_{ij}$ (continuous); (iii) max single-pair overlap (worst local hot-spot); (iv) overlap area per violation count (concentration of remaining overlaps).

R Independent classical force-directed legalizer (24 trials)

FD-pure = pairwise overlap repulsion + boundary force on all macros (no mask, no spring). **FD+spring** = + degree-normalised net-spring with weight 1.0, NTUplace3-style. Both operators run T=100 steps, step size 0.3, on the same 6 circuits $\times$ 4 seeds as Table 1, sharing the diffusion-sampled baseline draft for pairing.

Method	violation count	overlap area	max single-pair	area / count
baseline	36083	255061.4	431.49	3.820
VSR-post lambda=2	17944	82536.0	292.57	3.178
VSR-post lambda=8	17819	88767.3	342.59	3.135

Table 18: Cross-circuit medians. At λ=2 VSR halves both the count and the total overlap area; the max single-pair drops modestly (431 $\rightarrow$ 293). At λ=8 count and area are unchanged within noise (the λ knob trades macros-only HPWL, not legality magnitude §4.6). The remaining $\sim$ 18k violations are not concentrated in a few hot-spots: area-per-count drops only 3.82 $\rightarrow$ 3.13 (-18%), so VSR removes both *many tiny* overlaps and many average-sized overlaps in roughly equal proportion. This matches the spatial pattern: violators are still spread across the canvas (bbox ratio $\geq$ 1), so the residual is structural and not fixable by more aggressive local repair.

Circuit	seeds n	FD-pure $\Delta v\%$	FD-pure $\Delta h\%$	FD+spring $\Delta v\%$	FD+spring $\Delta h\%$	VSR-post (λ=8) $\Delta v\%$	VSR-post (λ=8) $\Delta h\%$
adaptec1	4	-36.3	$+184.8$	-36.0	$+35.8$	-35.4	-38.2
adaptec2	4	-56.8	$+258.1$	-52.1	$+53.9$	-51.6	-28.3
adaptec3	4	-51.5	$+138.7$	-51.5	$+40.9$	-52.9	-8.1
adaptec4	4	-61.9	$+121.3$	-59.7	$+9.7$	-55.6	-37.5
bigblue1	4	-20.1	$+148.6$	-22.6	-63.9	-23.5	-75.3
bigblue3	4	-74.5	$+109.1$	-75.7	$+31.7$	-70.2	-30.6
median	–	-54.2	$+143.6$	-51.8	$+33.8$	-52.2	-34.1

Table 19: Classical force-directed legalizer on the full 24 trials (6 circuits $\times$ 4 seeds, R4/Q5). Cross-circuit medians: FD-pure Δv=-54%/Δh=$+144\%$; FD+spring Δv=-52%/Δh=$+34\%$; VSR-post(λ=8) Δv=-52%/Δh=-34%. Comparable legality cut, $\sim$ 70pp better HPWL.

S LOCO cross-validation for λ

For each held-out circuit, choose $\hat{\lambda}$ that maximises the strict-Pareto count on the other 5 circuits, then evaluate on the held-out one.

T Intra-sampling timestep figure

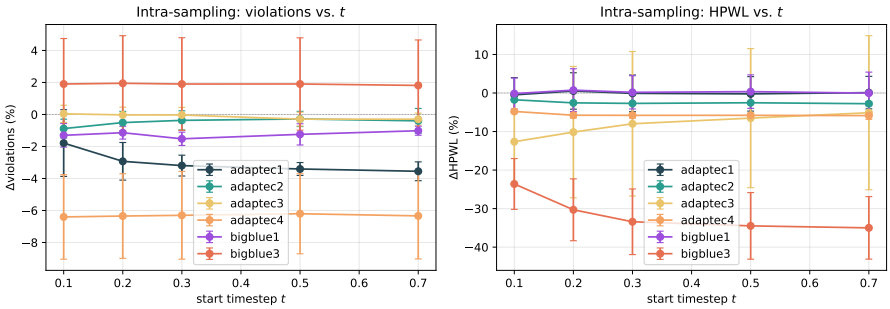

Figure 6: Intra-sampling HPWL change vs. t_{start}, 6 circuits $\times$ 3 seeds. Bigblue3 shows a striking monotonic improvement peaking at -34.5% HPWL.

U Reproducibility and released artifacts

All code, configs, and result JSONs are released. The main-paper numbers reproduce on a single A800 80 GB GPU (full sweep $\sim$ 70 min):

```
python scripts/run_main_neurips.py        # main result, ~30 min
python scripts/run_timestep_sweep.py      # timestep sweep, ~12 min
```

Circuit	seeds n	cg w=2 $\Delta v\%$	cg w=2 $\Delta h\%$	cg w=8 $\Delta v\%$	cg w=8 $\Delta h\%$	RePaint t=0.3 $\Delta v\%$	RePaint t=0.3 $\Delta h\%$	RePaint t=0.5 $\Delta v\%$	RePaint t=0.5 $\Delta h\%$
adaptec1	4	$+1.0$	$+52.9$	-3.1	$+57.3$	$+145.4$	-17.3	$+131.2$	-19.8
adaptec2	4	-1.6	$+27.7$	-0.6	$+15.2$	$+21.8$	-26.2	$+17.0$	-27.2
adaptec3	4	$+2.9$	$+12.1$	$+2.6$	$+11.7$	$+138.6$	-68.1	$+121.2$	-69.1
adaptec4	4	-0.7	$+4.0$	-0.6	$+1.3$	$+195.3$	-63.1	$+221.0$	-63.1
bigblue1	4	$+2.7$	$+6.2$	$+2.8$	$+1.5$	$+167.5$	-54.3	$+180.6$	-52.7
bigblue3	4	$+7.0$	$+11.7$	$+7.2$	$+11.8$	-41.8	-63.0	-12.8	-62.5

Table 20: cg ($w \in \{2, 8\}$) and RePaint ($t_{\text{start}} \in \{0.3, 0.5\}$) on the same 24 trials as Table 1 (R3/Q4). Replaces the earlier seed-42-only sweep at the two best per-circuit settings of each family.

held-out	$\hat{\lambda}$	test $\Delta v\%$	test $\Delta h\%$	strict-Pareto?	oracle $\lambda^\star$
adaptec1	8.0	-39.4	-34.4	✓	8.0
adaptec2	8.0	-48.9	-32.5	✓	8.0
adaptec3	8.0	-54.7	-13.6	✓	8.0
adaptec4	8.0	-54.9	-41.6	✓	8.0
bigblue1	8.0	-23.5	-72.5	✓	1.0
bigblue3	8.0	-70.1	-32.7	✓	8.0
summary	–	-51.8	-33.6	**6/6**	–

Table 21: LOCO selection on the λ-sweep. All 6 folds select $\hat{\lambda}$=8; held-out test gives **6/6 strict-Pareto improvement** matching the per-circuit oracle. Raw rows: `results/vsr_extra/lambda_loco.csv`.

```
python scripts/run_mem_profile.py         # memory profile, ~16 min
python scripts/run_component_ablation.py  # component ablation, ~12 min
python scripts/compute_neurips_stats.py
python scripts/analyze_extra_experiments.py
python scripts/analyze_component_ablation.py
```

Seeds: {42, 123, 300, 2024} for 4-seed runs; {42, 123, 300} for 3-seed sweeps. Result JSONs: `results/ispd2005/` and `results/vsr_extra/`.